C Programs – Slot D Assignment 1

1. Cab Fare Calculation

```
#include <stdio.h>

int main() {
    int distance, member;
    float fare = 100;

    printf("Distance: ");
    scanf("%d", &distance);
    printf("Member (1 = true, 0 = false): ");
    scanf("%d", &member);

    if (distance > 10)
        fare += 20 * (distance - 10);
    if (member)
        fare *= 0.9;

    printf("Final Fare: Rs.%g\n", fare);
    return 0;
}
```

2. Online Examination System

```
#include <stdio.h>

int main() {
    int answered, remaining, submitted;

    scanf("%d %d %d", &answered, &remaining, &submitted);

    if (answered >= 1 && remaining > 0 && submitted == 0)
        printf("Submission Allowed\n");
    else
        printf("Submission Not Allowed\n");
    return 0;
}
```

3. Treasure Hunt

```
#include <stdio.h>

int main() {
    int total, x, y;
    scanf("%d %d %d", &total, &x, &y);
```

```
    int ben = total * x / 100;
    int remaining = total - ben;
    int blackbeard = remaining * y / 100;
    remaining -= blackbeard;
    int each = remaining / 3;   // leftover coins are thrown into the sea

    printf("%d\n%d\n%d\n", ben, blackbeard, each);
    return 0;
}
```

4. Dollars & Cents

```
#include <stdio.h>

int main() {
    int d1, c1, d2, c2;
    scanf("%d %d %d %d", &d1, &c1, &d2, &c2);

    int cents = c1 + c2;
    int dollars = d1 + d2 + cents / 100;
    cents = cents % 100;

    printf("%d\n%d\n", dollars, cents);
    return 0;
}
```

5. Car Mileage

```
#include <stdio.h>

int main() {
    float mileage;
    int petrol, distance;
    scanf("%f %d %d", &mileage, &petrol, &distance);

    if (mileage * petrol >= distance)
        printf("Can reach\n");
    else
        printf("Cannot reach\n");
    return 0;
}
```

6. Car Race

```
#include <stdio.h>

int main() {
    int a, b, c, car;
    scanf("%d %d %d %d", &a, &b, &c, &car);

    if (a % car == 0)
```

```
        printf("Car 1 goes into road A\n");
    else if (b % car == 0)
        printf("Car 1 goes into road B\n");
    else if (c % car == 0)
        printf("Car 1 goes into road C\n");
    else
        printf("No path exists\n");
    return 0;
}
```

7. Debt Repay

```
#include <stdio.h>

int main() {
    int p, r, n;
    scanf("%d %d %d", &p, &r, &n);

    float interest = (float)p * r * n / 100;
    float amount = p + interest;
    float discount = interest * 0.02;
    float settlement = amount - discount;

    printf("%.2f\n%.2f\n%.2f\n%.2f\n", interest, amount, discount, settlement);
    return 0;
}
```

8. Employee Job Level

```
#include <stdio.h>

int main() {
    int exp, perf;
    printf("Experience: ");
    scanf("%d", &exp);
    printf("Performance: ");
    scanf("%d", &perf);

    if (exp < 3)
        printf("Level: Junior\n");
    else if (exp <= 7) {
        if (perf >= 80)
            printf("Level: Mid-Level\n");
        else
            printf("Level: Junior\n");
    } else {
        if (perf >= 85)
            printf("Level: Senior\n");
        else if (perf >= 80)
            printf("Level: Mid-Level\n");
        else
            printf("Level: Junior\n");
    }
```

```
    return 0;
}
```

9. Number of Days

```
#include <stdio.h>

int main() {
    int year, month, days;
    scanf("%d %d", &year, &month);

    if (year < 1900 || year > 9999) {
        printf("0\n");
        return 0;
    }

    switch (month) {
        case 1: case 3: case 5: case 7: case 8: case 10: case 12:
            days = 31; break;
        case 4: case 6: case 9: case 11:
            days = 30; break;
        case 2:
            if ((year % 4 == 0 && year % 100 != 0) || year % 400 == 0)
                days = 29;
            else
                days = 28;
            break;
        default:
            printf("0\n");
            return 0;
    }

    printf("%d Days\n", days);
    return 0;
}
```

10. Electricity Bill

```
#include <stdio.h>

int main() {
    int units;
    float bill;
    scanf("%d", &units);

    if (units <= 200)
        bill = units * 0.5;
    else if (units <= 400)
        bill = units * 0.65 + 100;
    else if (units <= 600)
        bill = units * 0.80 + 200;
    else
        bill = units * 1.25 + 425;
```

```
    printf("Rs.%g\n", bill);
    return 0;
}
```